# NIVAAR AI — MULTIMODAL PREDICTIVE MAINTENANCE REPORT

Predict Failures. Prevent Disruptions. | Triboelectric Sensing & Computer Vision Symbiosis

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|---|---|---|---|
| **Equipment Asset** | High-Voltage Induction Motor M-104 | **Asset Tag** | EQ-IND-M104 |
| **Category** | Industrial Motors | **Physical Location** | Petrochemical Unit 4, Compressor Line A |
| **Operating Status** | **CRITICAL** | **Symbiosis Health Index** | **48.2%** |
| **Inspection Timestamp** | 2026-10-23 08:28 UTC | **Multimodal Models** | TENG Isolation Forest + OpenCV Vision |

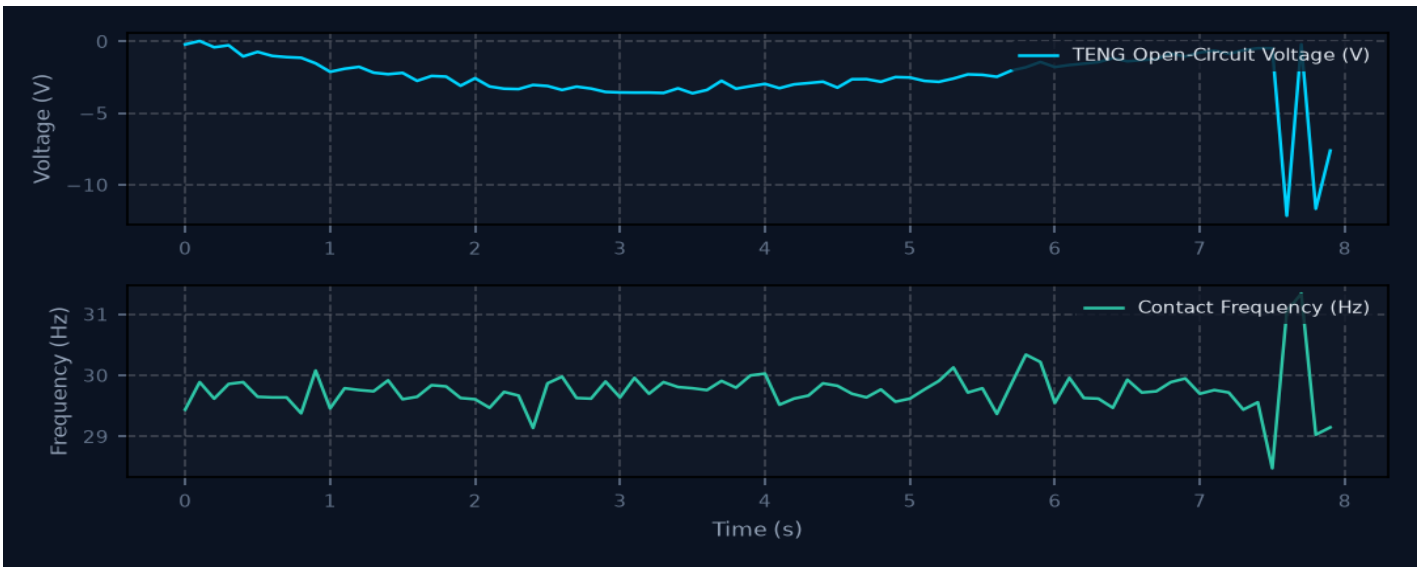
## 1. SYMBIOSIS — MULTIMODAL AI FUSION ASSESSMENT

**Primary Diagnostic Cause:** Coupled Structural Fatigue & Mechanical Impact Shock (Overall Severity: **CRITICAL**, Confidence: 82%). Multimodal synthesis fused TENG vibration telemetry, computer vision defect inspection, and operational asset history.

| Diagnostic Stream | Attribution Weight | Observed Diagnostic Finding |
|---|---|---|
| TENG Sensor Telemetry | 75.0% | Impact kurtosis and open-circuit voltage spikes (Vpp) |
| Computer Vision Inspection | 0.0% | Bounding box localized crack/corrosion surface area |
| Machine Operational History | 25.0% | Baseline compliance and previous maintenance tickets |

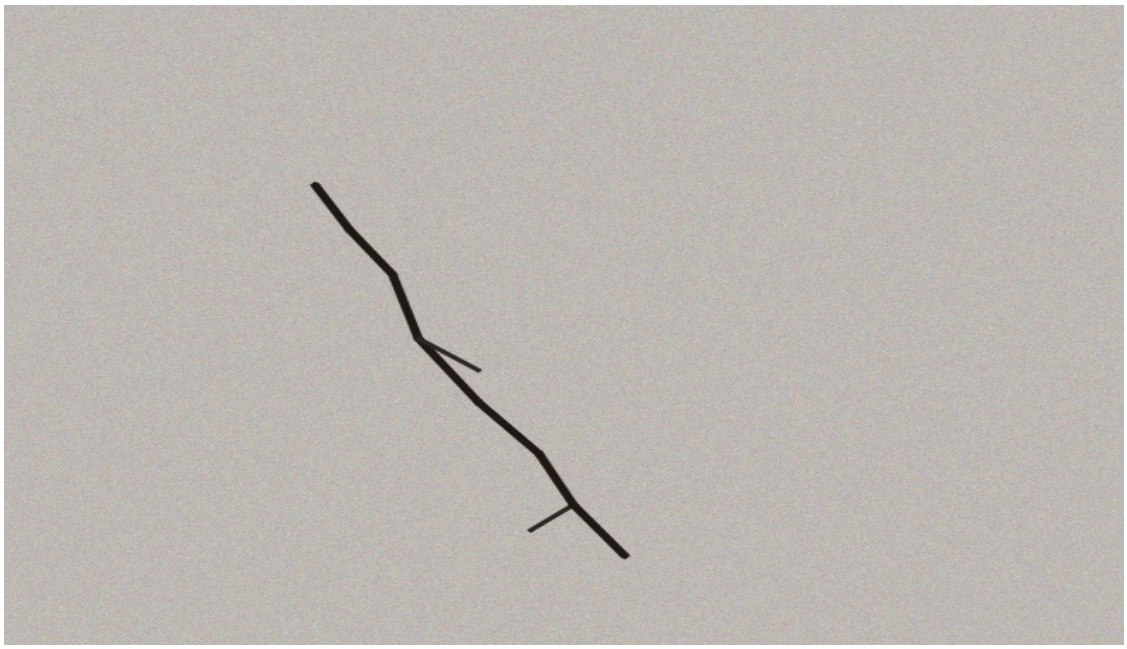
## 2. TENG (TRIBOELECTRIC NANOGENERATOR) TELEMETRY & PHYSICAL UNITS

| Metric Parameter | Measured Value | Preserved Physical Unit | Baseline Compliance |
|---|---|---|---|
| Peak-to-Peak Voltage (V_pp) | 16.80 | Volts (V) | Nominal <= 12.0 V |
| Root Mean Square Voltage (V_rms) | 4.12 | Volts (V) | Nominal <= 3.5 V |
| Dominant Vibration Frequency | 29.75 | Hertz (Hz) | Running Speed 1X |
| Impulsive Kurtosis (Impact Shock) | 4.85 | Dimensionless ratio | Nominal ~ 3.0 |
| Average Charge Transfer per Cycle | 4.20 | Nanocoulombs (nC) | Expected 2.0 - 5.0 nC |




## 3. COMPUTER VISION DEFECT IDENTIFICATION & ANNOTATED EVIDENCE

**Reference Component Visual Baseline:**



## 4. RECOMMENDED ENGINEERING INSPECTION ACTIONS

• Priority 1: Conduct on-site dye penetrant / ultrasound NDT testing across flagged crack coordinates.
• Priority 2: Perform ultrasound grease listening on bearings to verify TENG impact signature.
• Priority 3: Re-measure laser coupling alignment before next operational shift.
• Advisory Notice: Multimodal fusion outputs are advisory decision-support findings; a qualified engineer must authorize mechanical work.

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***DATA PROVENANCE, METHODOLOGY & REGULATORY DISCLAIMER:***
*Nivaar AI evaluates real-time TENG triboelectric vibration signals via unsupervised Isolation Forest and extracts structural defect bounding boxes using calibrated OpenCV computer vision morphology. All statistics, voltage units (V), frequencies (Hz), and pixel coordinates are computed directly from ingested source data without synthetic inflation. This document is an engineering decision-support tool and does not constitute a certified destructive physical teardown report. Qualified engineering personnel must perform physical verification prior to taking machinery offline.*